

Institute of Actuaries of India

May 2010 EXAMINATION

**Subject ST6 — Finance and Investment B
Specialist Technical**

Indicative Solution

- 1(a) The firm should enter a swap in which the firm pays a 8% fixed rate and receives MIBOR on Rs. 100 crores of notional principal. Its total payments will be as follows:

Interest payments on bond	$(\text{MIBOR} + 0.005) \times \text{Rs. 100 crores par value}$
<u>Net cash flow from swap</u>	<u>$(0.08 - \text{MIBOR}) \times \text{Rs. 100 crores notional principal}$</u>
Total	$0.085 \times \text{Rs. 100 crores}$

The interest rate on the synthetic fixed-rate loan is 8.5%.

- 1(b) The firm receives a fixed rate that is 2% higher than the market rate. The extra payment of $0.02 \times \text{Rs. 100 crores}$ has present value equal to:

$$\text{Rs. } 20,000,000 \times \text{Annuity factor}(6\%, 3) = \text{Rs. } 53,460,239$$

$$1(c) \text{ Value of the bond} = \sum_{t=1}^T \frac{c}{(1+y)^t} + \frac{1}{(1+y)^T} = \frac{c[(1+y)^T - 1] + y}{y(1+y)^T}$$

$$\text{Duration of the bond} = D = \frac{\sum_{t=0}^T t \frac{c}{(1+y)^t} + \frac{T}{(1+y)^T}}{\frac{c[(1+y)^T - 1] + y}{y(1+y)^T}}$$

Assume:

$$X = \sum_{t=1}^T t \frac{c}{(1+y)^t} + \frac{T}{(1+y)^T} = c \left[\frac{1}{1+y} + \frac{2}{(1+y)^2} + \frac{3}{(1+y)^3} + \dots + \frac{T}{(1+y)^T} \right] + \frac{T}{(1+y)^T}$$

$$X(1+y) = c \left[1 + \frac{2}{(1+y)^1} + \frac{3}{(1+y)^2} + \frac{4}{(1+y)^3} + \dots + \frac{T}{(1+y)^{T-1}} \right] + \frac{T}{(1+y)^{T-1}}$$

$$X(1+y) - X = c + c \left[\frac{1}{(1+y)^1} + \frac{1}{(1+y)^2} + \frac{1}{(1+y)^3} + \dots + \frac{1}{(1+y)^{T-1}} \right] + \frac{T}{(1+y)^{T-1}} - \frac{T(c+1)}{(1+y)^T}$$

$$Xy = \sum_{t=1}^T \frac{c}{(1+y)^t} + \frac{1}{(1+y)^T} - \frac{c+1}{(1+y)^T} + c + \frac{T}{(1+y)^{T-1}} - \frac{T(c+1)}{(1+y)^T}$$

$$Xy = \frac{c[(1+y)^T - 1] + y}{y(1+y)^T} + \frac{c[(1+y)^T - 1] - 1 - T(c-y) + y - y}{(1+y)^T}$$

$$Xy = \frac{[c[(1+y)^T - 1] + y](1+y) - [(1+y) - T(c-y)]y}{y(1+y)^T}$$

$$X = \frac{1}{y} \left[\frac{[c[(1+y)^T - 1] + y](1+y) - [(1+y) - T(c-y)]y}{y(1+y)^T} \right]$$

$$D = \frac{1+y}{y} - \frac{(1+y) + T(c-y)}{c[(1+y)^T - 1] + y}$$

- 1(d) (i) PV of obligation = Rs. 120 crores/0.12 = Rs. 1000 crores
Duration of obligation = 1.12/0.12 = 9.33 years

$$\text{Duration of 6-year 10\% bond} = \frac{1.12}{0.12} - \frac{1.12 + 6(0.10 - 0.12)}{0.10(1.12^6 - 1) + 0.12} = 4.73 \text{ years}$$

$$\text{Duration of 21-year 5\% bond} = \frac{1.12}{0.12} - \frac{1.12 + 21(0.05 - 0.12)}{0.05(1.12^{21} - 1) + 0.12} = 9.91 \text{ years}$$

Call w the weight on the 6-year maturity bond (which has duration of 4.73 years). Then:

$$(w \times 4.73) + [(1 - w) \times 9.91] = 9.33 \Rightarrow w = 0.1109$$

Therefore: $0.1109 \times \text{Rs. 1000 crores} = \text{Rs. 110.8647 crores}$ in the 6-year bond, and;

$$0.8891 \times \text{Rs. 1000 crores} = \text{Rs. 889.1353 crores}$$
 in the 21-year bond.

(ii) The price of the 21-year bond is:

$$[\text{Rs. } 50 \times \text{Annuity factor (12\%, 21)}] + [\text{Rs. } 1,000 \times \text{PV factor (12\%, 21)}] \\ = \text{Rs. 470.6598}$$

Therefore, the bond sells for 0.4707 times its par value, and:

$$\text{Market value} = \text{Par value} \times 0.4707$$

$$\text{Rs. 889.1353 crores} = \text{Par value} \times 0.4707 \Rightarrow \text{Par value} = \text{Rs. 1889.125 crores.}$$

The price of the 6-year bond is:

$$[\text{Rs. } 100 \times \text{Annuity factor (12\%, 6)}] + [\text{Rs. } 1,000 \times \text{PV factor (12\%, 6)}] = \text{Rs. } 917.7719$$

Therefore, the bond sells for 0.9178 times its par value, and:

$$\text{Market value} = \text{Par value} \times 0.9178$$

$$\text{Rs. 110.8647 crores} = \text{Par value} \times 0.9178 \Rightarrow \text{Par value} = \text{Rs. 120.7976 crores.}$$

[13]

$$\begin{aligned} 2(a) \quad V(0.99, 1) &= F \frac{\alpha F^{-1}(0.02) + \sqrt{0.25} F^{-1}(0.99)}{\sqrt{1 - 0.25}} \\ &= F \left(\frac{-2.0538 + 0.5 \times 2.3263}{0.8660} \right) = F(-1.0284) = 0.1519 \end{aligned}$$

The 1-year 99% credit VaR is therefore:

$$4000 \times 0.1519 \times (1 - 0.5) = \text{Rs. 303.7864 crores}$$

2(b)

To make the portfolio gamma and vega neutral, the two traded options on the portfolio can be used. Assume x and y are the quantities of option 1 and option 2 are added to the portfolio. We require that:

$$-4100 + 1.20x + 0.80y = 0$$

$$-2100 + 0.60x + 0.30y = 0$$

The solution to these equations is $x = 3750$, $y = -500$.

The portfolio can thus be made gamma neutral and vega neutral by purchasing 3750 of Option 1 and by selling 500 of Option 2.

The delta of the portfolio, after the addition of the positions in the two traded options is:

$$700 + 3750 \times 0.60 + (-500) \times 0.30 = 2800$$

Hence 2800 units of the asset would have to be sold to maintain delta neutrality.

[7]

3.

Number of days between July 1 2008 and November 30 2008 = 152

Number of days between November 30 2008 and January 1 2009 = 32

Number of days between November 30 2008 and July 1 2009 = $32 + 181 = 213$

Number of days between November 30 2008 and January 1 2010 = $32 + 181 + 184 = 397$

Number of days between November 30 2008 and March 31 2010 = $32 + 181 + 184 + 89 = 486$.

The last coupon date was 152 days ago. The current quoted bond price is Rs. 120. The cash price of the bond is obtained by adding to the quoted price the proportion of the next payment that accrues to the holder. The cash price is therefore:

$$120 + \frac{152}{152 + 32} \times 5 = \text{Rs. } 124.1304$$

Three coupon payments of Rs. 5 each will be received after 32 days (= 0.0877 years), 213 days (= 0.5836 years) and 397 days (= 1.0877 years) from today (November 30 2008). The present value of these coupon payments is:

$$5(e^{-0.08 \times 0.0877} + e^{-0.08 \times 0.5836} + e^{-0.08 \times 1.0877}) = 5 \times 2.8641 = \text{Rs. } 14.3203.$$

The futures contract lasts for 486 days (= 1.3315 years). The cash futures price if the contract were written on the 10% bond, would therefore be:

$$(124.1304 - 14.3203)e^{0.08 \times 1.3315} = \text{Rs. } 122.1528$$

At delivery, there are 89 days (from January 1 2010 to March 31 2010) of accrued interest. There are 92 days from March 31 2010 to July 1 2010). The quoted futures price, if the contract were written on the 10% coupon bond, is calculated by subtracting the accrued interest:

$$122.1528 - \frac{89}{89 + 92} \times 5 = 119.6942$$

From the definition of the conversion factor, 1.3200 standard bonds are considered equivalent to each 10% coupon bond. The quoted futures price should therefore be:

$$\frac{119.6942}{1.3200} = 90.6774 .$$

[7]

4(a)

From put-call parity:

$$C - P = S_0 - K/(1 + r_f)^T$$

If the options are at the money, then $S_0 = K$ and:

$$C - P = K - K/(1 + r_f)^T$$

The right-hand side of the equation is positive, and we conclude that $C > P$.

4(b)

The farmer has the option to sell the crop to the government for a guaranteed minimum price if the market price is too low.

If the support price is denoted P_S and the market price P_m then the farmer has a put option to sell the crop (the asset) at an exercise price of P_S even if the price of the underlying asset (P) is less than P_S .

4(c)

The manager gets a bonus if the stock price exceeds a certain value and gets nothing otherwise.

This is the same as the payoff to a call option.

4(d)

A straddle is a call and a put.

The delta of the straddle is the sum of the deltas of the individual options:

$$0.5 + (-0.7) = -0.2$$

4(e)

A \$1 increase in a call option's exercise price would lead to a decrease in the option's value of less than \$1.

The change in the call price would equal \$1 only if: (i) there were a 100% probability that the call would be exercised, and (ii) the interest rate were zero.

4(f)

Implied volatility has increased.

If not, the call price would have fallen as a result of the decrease in stock price.

4(g)

Short selling results in an immediate cash inflow, whereas the short futures position does not:

Action	Initial CF	Final CF
Short Sale	$+P_0$	$-S_T$
Short Futures	0	$F_0 - S_T$

4(h)

(i). False.

The parity relationship tells us that the futures price is determined by the stock price, the interest rate, and the dividend yield; it is not a function of beta.

(ii). True.

The short futures position will profit when the market falls. This is a negative beta position.

[12]

5.

a. $5000 \times 1.03 = 5150$

b. $\text{Rs. } 100 \text{ crores} \times 1.10 / (50 \times 5000) = 4400 \text{ contracts short}$

c. $4400 \times 50 \times (5150 - S_T) = 1,133,000,000 - 220,000S_T$

d. The expected return on a stock is:

$$\alpha + r_f + \beta[E(r_M) - r_f]$$

The CAPM predicts that $\alpha = 0$. In this case, however, if you believe that $\alpha = 3\%$ (i.e., 0.03), then you forecast a portfolio return of:

$$r_P = 0.03 + 0.03 + 1.10 \times (r_M - 0.03) + \varepsilon = 0.06 + [1.10 \times (r_M - 0.03)] + \varepsilon$$

where ε is diversifiable risk, with an expected value of zero.

e. Because the market is assumed to pay no dividends:

$$r_M (\text{per six months}) = (S_T - 5000) / 5000 = (S_T / 5000) - 1$$

where S_T is the value of Nifty in six months

The rate of return can also be written as:

$$r_P = 0.06 + (1.10 \times [(S_T / 5000) - 1 - 0.03]) + \varepsilon$$

The rupee value of the stock portfolio as a function of the market index is therefore:

$$\text{Rs. } 100 \text{ crores} \times (1 + r_P) =$$

$$\text{Rs. } 100 \text{ crores} \times [1.10 \times (S_T / 5000) - 0.073 + \varepsilon] =$$

$$-\text{Rs. } 73,000,000 + 220,000S_T + (\text{Rs. } 100 \text{ crores} \times \epsilon)$$

The rupee value of the short futures position will be (from part c):

$$1,133,000,000 - 220,000S_T$$

The total value of the portfolio plus the futures proceeds is therefore:

$$[-73,000,000 + 220,000S_T + (100 \text{ crores} \times \epsilon)] + [1,133,000,000 - 220,000S_T] =$$

$$\text{Rs. } 1,060,000,000 + (\text{Rs. } 100 \text{ crores} \times \epsilon)$$

The payoff is independent of the value of the stock index. Systematic risk has been eliminated by hedging (although firm-specific risk remains).

- f. The portfolio-plus-futures position cost Rs. 100 crores to establish.

The expected end-of-period value is Rs. 106 crores.

The rate of return is therefore 6%.

- g. The beta of the hedged position is 0.

The fair return should be $r_f = 3\%$. Therefore, the alpha of the position is $(6\% - 3\%) = 3\%$, the same as the alpha of the portfolio. Now, however, one can take a position on the alpha without incurring systematic risk.

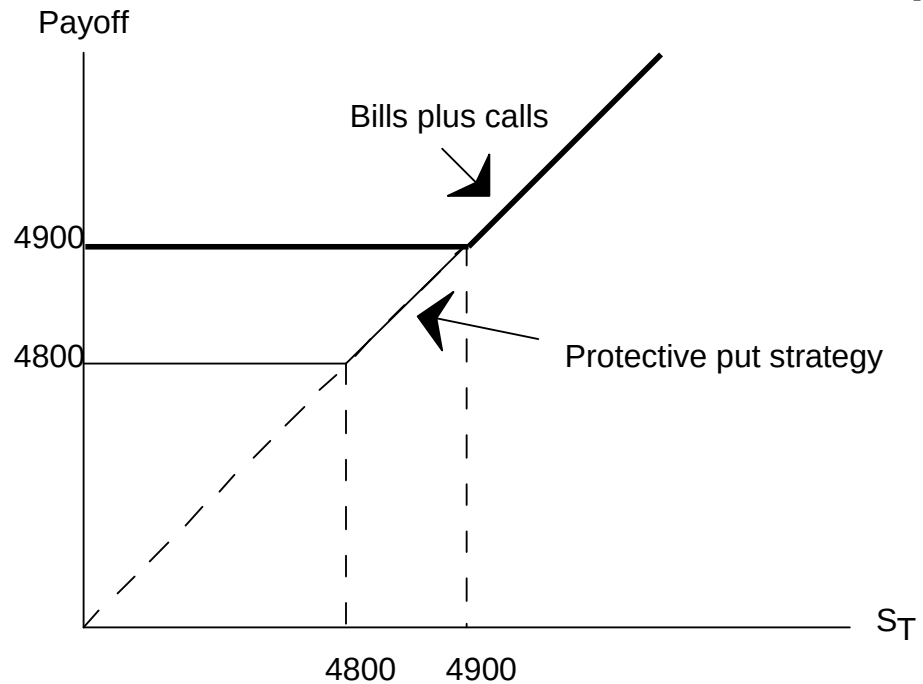
[11]

6.

a.

Position	$S_T \leq 4800$	$S_T > 4800$
Buy stock	S_T	S_T
Buy put	$4800 - S_T$	0
Total	4800	S_T

Position	$S_T \leq 4900$	$S_T > 4900$
Buy call	0	$S_T - 4900$
Buy T-bills	4900	4900
Total	4900	S_T



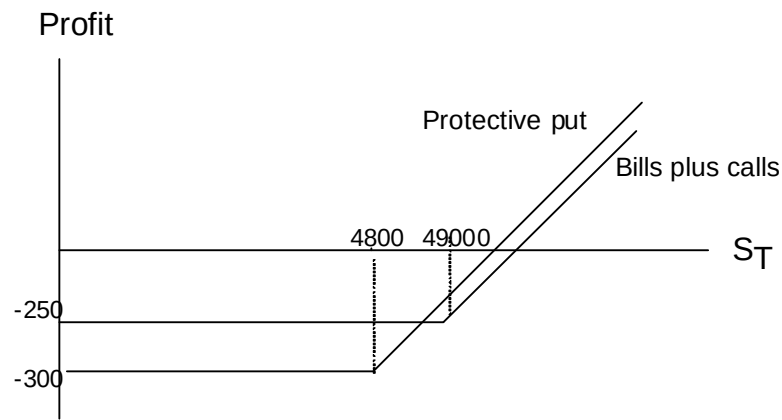
- b. The bills plus call strategy has a greater payoff for some values of S_T and never a lower payoff.

Since its payoffs are always at least as attractive and sometimes greater, it must be more costly to purchase.

- c. The initial cost of the stock plus put position is: $5000 + 100 = \text{Rs. } 5100$

The initial cost of the bills plus call position is: $4850 + 300 = \text{Rs. } 5150$

	$S_T = 4500$	$S_T = 4800$	$S_T = 4900$	$S_T = 5000$	$S_T = 5100$	$S_T = 5400$
Stock	4500	4800	4900	5000	5100	5400
+ Put	300	100	0	0	0	0
Payoff	4800	4800	4900	5000	5100	5400
Profit	-300	-300	-200	-100	0	300
Bill	4900	4900	4900	4900	4900	4900
+ Call	0	0	0	100	200	500
Payoff	4900	4900	4900	5000	5100	5400
Profit	-250	-250	-250	-150	-50	250



- d. The stock and put strategy is riskier.

This strategy performs worse when the market is down and better when the market is up. Therefore, its beta is higher.

- e. Parity is not violated because these options have different exercise prices. Parity applies only to puts and calls with the same exercise price and expiration date.

[9]

7(a)

$$P(t, T) = e^{-\int_t^T \partial f(t, u) du}$$

Thus, we have:

$$P(t, T) = e^{X(t, T)}$$

$$\text{where } X(t, T) = -\int_t^T \partial f(t, u) du$$

$$dX(t, T) = f(t, t) - \int_t^T \partial f(t, u) du$$

$$dX(t, T) = r(t)dt - \int_t^T [m(t, u)dt + s(t, u)dz(t)]du$$

$$dX(t, T) = [r(t) - \int_t^T m(t, u)du]dt - [s(t, u)du]dz(t)$$

Using Ito's formula $P(t, T) = e^{X(t, T)}$, we get:

$$dP(t, T) = \{r(t) + b(t, T)\}P(t, T)dt + a(t, T)P(t, T)dz(t)$$

Where

$$\begin{aligned} \dot{a}(t, T) &= -\int_t^T \sigma(t, u) du \\ \dot{b}(t, T) &= -\int_t^T m(t, u) du + \frac{1}{2} \sigma^2(t, T) \end{aligned}$$

7(b)

The drift of any process representing the value of a tradeable asset under the risk neutral martingale measure is equal to the short rate. This means that $b(t, T) = 0$.

Using the expression for $b(t, T)$ and taking the derivative with respect to T we get the HJM drift condition:

$$m(t, T) = s(t, T) \int_t^T \sigma(t, u) du$$

[7]

8.

The price of the pay later option is given by:

$$P_t = e^{-r_d(T-t)} E^d[(E_T - K - P) I\{E_T > K\} | F_t]$$

where the super-index d on the expectation indicates that the expectation should be taken under the domestic martingale measure Q^d

$$P_t = e^{-r_d(T-t)} E^d[(E_T - K) I\{E_T > K\} | F_t] - P E^d[I\{E_T > K\} | F_t]$$

The dynamics of E under Q^d are given by:

$$dE = (r_d - r_f) E dt + S_E E dW^d$$

EB^f / B^d is a Q^d martingale. Thus W^d is a Q^d wiener process

$$P_t = e^{-r_f(T-t)} e^{-(r_d - r_f)(T-t)} E^d[(E_T - K) I\{E_T > K\} | F_t] - e^{-r_d(T-t)} P Q^d(X_T > K)$$

$$P_t = e^{-r_f(T-t)} \{E_t F[d_1(t)] - e^{-(r_d - r_f)(T-t)} K F[d_2(t)]\} - e^{-r_d(T-t)} P Q^d(X_T > K)$$

$$\text{where } d_1(t) = \frac{1}{S_E \sqrt{T-t}} \left\{ \ln\left(\frac{E_t}{K}\right) + (r_d - r_f + \frac{1}{2} S_E^2)(T-t) \right\}$$

$$d_2(t) = d_1(t) - S_E \sqrt{T-t}$$

Since E is a GBM it has the solution:

$$E_T = E_t \exp \left\{ \left(r_d - r_f - \frac{1}{2} s_E^2 \right) (T - t) + s_E (W_T^d - W_t^d) \right\}$$

$$E_T = E_t e^Z, \text{ where } Z \sim N \left((r_d - r_f - s_E^2 / 2)(T - t), s_E^2 (T - t) \right).$$

Thus, the probability in the second term is:

$$Q^d(E_T > K) = F[d_2(t)]$$

The price of the pay later option at time t is thus:

$$P_t = e^{-r_f(T-t)} E_t F[d_1(t)] - e^{-r_d(T-t)} K F[d_2(t)] - e^{-r_d(T-t)} P F[d_2(t)]$$

By putting $t = 0$ and $P_0 = 0$, the premium P is given by:

$$P = e^{(r_d - r_f)T} E_0 \frac{F[d_1(0)]}{F[d_2(0)]} - K$$

[12]

9(a).

The payoff function describing the option is given by:

$$g(S_T) = \text{Max}(S_T, K)$$

$$\text{Max}(S_T, K) = K + \text{Max}(S_T - K, 0)$$

Thus, the price P of the option at time t is given as:

$$P_t = e^{-r(T-t)} E^Q[K + \text{Max}(S_T - K, 0) | F_t] = e^{-r(T-t)} K + c(t, S_t, K, T, r, s)$$

where $c(t, S_t, K, T, r, s)$ denotes the standard Black-Scholes price at time t of a European call option with exercise price K and expiry date is T, when the price of the underlying at time t is S_t , the risk-free interest rate is r, and the volatility of the underlying is s .

$$P_t = e^{-r(T-t)} K + S_t F[d_1(t, S_t)] - e^{-r(T-t)} K F[d_2(t, S_t)]$$

Using $F(-x) = 1 - F(x)$, we get:

$$P_t = S_t F[d_1(t, S_t)] + e^{-r(T-t)} K F[-d_2(t, S_t)]$$

9(b)

The price of the option at time $t \in [0, T_0]$ is given by:

$$P_t = e^{-r(T-t)} E^Q[\text{Max}(S_T, S_{T_0}) | F_t] = e^{-r(T-t)} E^Q[E^Q[\text{Max}(S_T, S_{T_0}) | F_{T_0}] | F_t]$$

Using $S_T = S_{T_0} \exp \left[\left(r - \frac{s^2}{2} \right) (T - T_0) + s (W_T - W_{T_0}) \right]$, we get:

$$P_t = e^{-r(T-t)} E^Q \left[\frac{S_{T_0}}{e^{-r(T-T_0)}} \max(e^{(r-\frac{s^2}{2})(T-T_0)+s(W_T-W_{T_0})}, 1) \mid F_{T_0} \right] \mid F_t$$

$$P_t = e^{-r(T-t)} E^Q \left[\frac{S_{T_0}}{e^{-r(T-T_0)}} \max(e^{(r-\frac{s^2}{2})(T-T_0)+s(W_T-W_{T_0})}, 1) \mid F_{T_0} \right] \mid F_t$$

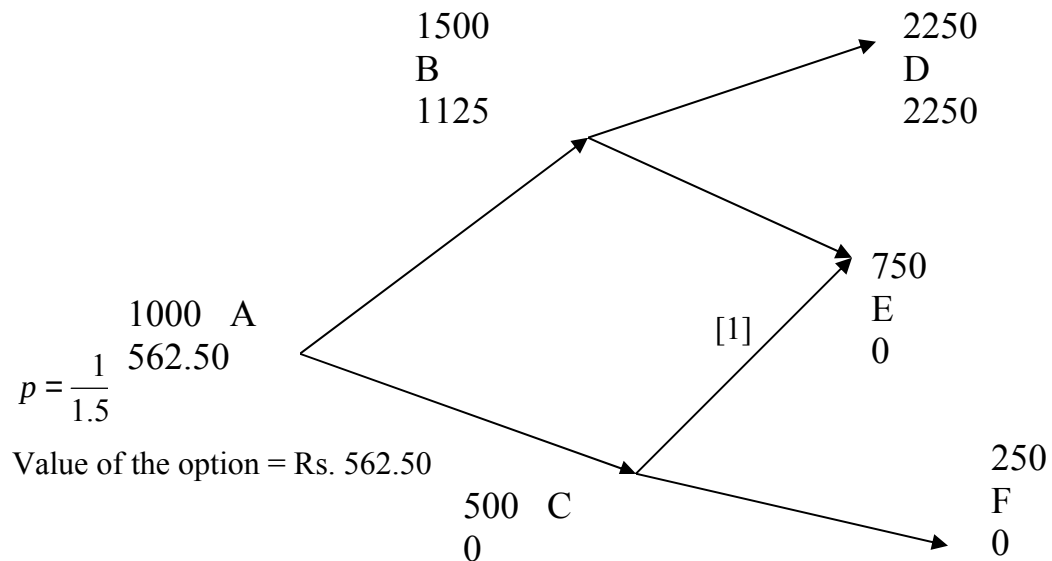
$$P_t = e^{rt} E^Q \left[\frac{S_{T_0}}{e^{rT_0}} P^a(T_0, 1, 1, T, r, s) \mid F_t \right]$$

$$P_t = S_t P^a(T_0, 1, 1, T, r, s) = S_t [e^{-r(T-T_0)} F[-d_2(T_0, 1)] + F[d_1(T_0, 1)]]$$

$P^a(T_0, 1, 1, T, r, s)$ denotes the price of the option at time T_0 in part (a), when the price of the underlying is 1, exercise price is 1, expiry date is T , risk-free interest rate is r , and the volatility of the underlying is s .

[9]

10(a)



(b)

To obtain the replicating portfolio at a one has to solve the following set of equations:

$$1500f + y = 1125$$

$$500f + y = 0$$

$$f = \frac{1125}{1000} = \frac{9}{8}; y = -562.50$$

To obtain the replicating portfolio at B one has to solve the following set of equations:

$$2250f + y = 2250$$

$$750f + y = 0$$

$$f = \frac{3}{2}; y = -1125$$

To obtain the replicating portfolio at C one has to solve the following set of equations:

$$750f + y = 0$$

$$250f + y = 0$$

$$f = 0; y = 0$$

That the portfolio is self financing is seen from the following equation

$$-562.50 + \frac{9}{8} \times 1500 = -1125 + \frac{3}{2} \times 1500 = 1125$$

$$-562.50 + \frac{9}{8} \times 500 = 0 + 0 \times 500 = 0$$

[6]

11(a)

$$r_t = f(t, t) = X_t + U(0) = X_t$$

11(b)

Using the relation $r_t = X_t$, we obtain:

$$df(t, T) = [a(t, r_t) - u(T - t)]dt + s(t, r_t)dW_t$$

where $u(y) = U'(y)$

As per HJM drift condition, we have:

$$a(t, r_t) - u(T - t) = \int_t^T s(t, r_t) \sigma(s, r_s) ds$$

$$a(t, r) - u(T - t) = s^2(t, r)(T - t)$$

Denoting $T - t$ by y this reads as:

$$a(t, r) - u(y) = s^2(t, r)y$$

Putting $y = 0$, we get:

$$a(t, r) = c$$

where $c = u(0)$.

Thus, we have:

$$c - u(y) = s^2(t, r)y$$

Which implies that s also has to be constant as a function of (t, r) .

$$u(y) = c - s^2 y$$

By integrating, we obtain:

$$U(y) = -\frac{s^2}{2}y + cy + a$$

[7]

[Total Marks 100]
